Status	Finished
Started	Monday, 26 October 2026, 1:25 AM
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Grade	**13.00** out of 15.00 (**86.67**%)

Question **1**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

- a. It is based on the divide-and-conquer approach
- b. It is less efficient than bubble sort
- c. It does not require recursion
- d. It sorts data using a single pass

The correct answer is: It is based on the divide-and-conquer approach

Question **2**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

- a. Bubble Sort does not guarantee sorted order
- b. Bubble Sort is difficult to implement
- c. Bubble Sort cannot handle duplicate elements
- d. Bubble Sort is less efficient for large lists

The correct answer is: Bubble Sort is less efficient for large lists

Question **3**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

- a. It divides the list into two halves, sorts each half, and then merges them
- b. It compares adjacent elements and swaps them if necessary
- c. It builds a sorted array one element at a time
- d. It repeatedly finds the minimum element and moves it to the sorted part of the list

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **4**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

a. It avoids the need for recursion
b. It simplifies the sorting process by using only one pass
c. It allows for efficient parallel processing and sorting of data
d. It only works on small datasets

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **5**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

a. Heap sort
b. Quick sort
c. Selection sort
d. Merge sort

The correct answer is:
Merge sort

Question **6**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

a. Bubble Sort
b. Quick Sort
c. Heap Sort
d. Merge Sort

The correct answer is: Bubble Sort

Question **7**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

- a. sorted()
- b. sort()
- c. order()
- d. arrange()

The correct answer is: sorted()

Question **8**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

- a. 80,150,200,470,40,90,300,400,70,120
- b. 40,70,80,90,120,150,200,300,400,470
- c. 40,80,90,150,200,300,400,470,70,120
- d. 200,470,80,150,40,90,300,400,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **9**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

- a. Provides a basis for other algorithms to work efficiently
- b. Makes data harder to manage
- c. Makes code execution slower
- d. Decreases the efficiency of algorithms

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **10**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

a. The parts are merged without sorting
b. Each part is searched for a specific element
c. The parts are ignored
d. Each part is sorted independently

The correct answer is: Each part is sorted independently

Question **11**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

a. Selection
b. All of the above
c. Searching
d. Duplicates

The correct answer is: All of the above

Question **12**

Complete

Mark 0.00 out of 1.00

Very slow way of sorting is_____

a. Bubble sort
b. Insertion sort
c. Heap sort
d. Quick sort

The correct answer is:
Insertion sort

Question **13**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

a. It has no effect on the analysis process
b. It makes the analysis slower
c. It allows for quicker identification of duplicates
d. It complicates the analysis process

The correct answer is: It allows for quicker identification of duplicates

Question **14**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

a. Finding a specific element in a list
b. Arranging data in a specific order
c. Merging two datasets
d. Removing duplicates from a list

The correct answer is: Arranging data in a specific order

Question **15**

Complete

Mark 0.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

a. Bubble Sort
b. Linear Search
c. Merge Sort
d. Quick Sort

The correct answer is: Merge Sort